

Business Requirements Document (BRD)

UC11: Enterprise Pharmaceutical Data Platform (EPDP)

Project: Building a Unified Pharmaceutical Manufacturing & Clinical Analytics Platform

Organization: ABC Pharmaceuticals Ltd.

Document Type: Business Requirements Document

Phase: Sprint 0 — Project Initiation & Architecture

Primary ETL Tool: Pentaho Data Integration (Spoon)

Database: PostgreSQL

Analytics: Power BI

Programming: Python (Pandas)

1. Introduction

ABC Pharmaceuticals Ltd. is a global pharmaceutical company involved in the research, manufacturing, and distribution of prescription medicines, vaccines, and healthcare products.

The organization generates large volumes of data from multiple independent enterprise systems, including ERP, LIMS, MES, QMS, CTMS, WMS, supply-chain systems, regulatory systems, and IoT-enabled manufacturing equipment.

The **Enterprise Pharmaceutical Data Platform (EPDP)** will provide a unified data engineering platform for collecting, cleaning, transforming, storing, governing, and analyzing pharmaceutical data.

2. Business Problem

The organization's data is generated across multiple independent systems. Because these systems operate separately, the organization faces challenges in:

- Maintaining pharmaceutical product and batch traceability
- Monitoring manufacturing quality
- Ensuring regulatory compliance
- Managing inventory efficiently
- Integrating laboratory and manufacturing information
- Supporting clinical research reporting

- Producing consistent management and regulatory reports

3. Business Objectives

The primary objectives of the EPDP are:

1. Collect pharmaceutical operational data from multiple heterogeneous enterprise systems.
2. Clean, validate, and standardize manufacturing, laboratory, and clinical data.
3. Store curated datasets in an enterprise PostgreSQL data warehouse.
4. Maintain metadata, audit information, and end-to-end data lineage.
5. Provide data for executive dashboards and business analytics.
6. Use Git and GitHub for collaborative version control.

4. Project Scope

The project will cover:

- Pharmaceutical data-source identification
- Data ingestion
- CSV, Excel, JSON, XML, and SQL data integration
- Data cleaning and validation
- Data profiling
- Data standardization
- PostgreSQL staging and data warehouse development
- ETL logging and exception handling
- Power BI dashboard development
- Git/GitHub-based version control

5. Stakeholders

The major stakeholders identified for the project are:

Stakeholder	Responsibility
Manufacturing Team	Production and batch-related data
Quality Assurance Team	Quality and compliance information
Laboratory Team	Laboratory and testing data
Clinical Research Team	Clinical research information

Regulatory Affairs Team	Regulatory and submission information
Supply Chain Team	Inventory and distribution information
Executive Management	Business insights and executive reporting
Data Engineering Team	Data ingestion, transformation and warehouse
Analytics Team	Data analysis and dashboards

6. Data Sources

The platform will integrate data from multiple pharmaceutical sources.

Source	Example Data
ERP	Product, supplier and business information
MES	Manufacturing and production data
LIMS	Laboratory test results
QMS	Quality information
CTMS	Clinical trial information
WMS	Warehouse and inventory information
Batch Production Records	Manufacturing batch information
Drug Stability Testing Reports	Drug stability and quality measurements
Regulatory Submission Files	Regulatory application
IoT Manufacturing Equipment Sensors	Equipment and sensor measurements

7. Functional Requirements

FR-01: Data Source Management

The system shall identify and document all pharmaceutical data sources used by the project.

The source inventory shall contain:

- Source-system name, Data source type, File/database format, Dataset description, Relevant fields, Expected data volume.

FR-02: Data Ingestion

The system shall ingest data from heterogeneous sources.

The ETL platform shall support:

- CSV, Excel, JSON, XML, SQL/database sources.

Pentaho Data Integration will be used to develop the ingestion pipelines.

FR-03: Data Cleaning

The system shall identify and handle common data-quality problems, including:

- Missing values, Duplicate records, Invalid values, Incorrect data types, Inconsistent formats, Inconsistent naming, Invalid dates, Out-of-range sensor measurements.

FR-04: Data Validation

The system shall validate incoming records before they are loaded into curated PostgreSQL tables.

Validation shall include:

- Required-field validation, Data-type validation, Range validation, Duplicate detection, Date validation.

FR-05: Data Standardization

The system shall standardize pharmaceutical data to provide consistent values across different datasets.

Examples include:

- Standardizing product names, Standardizing supplier names, Standardizing units, Standardizing dates, Standardizing batch identifiers, Standardizing laboratory values, Standardizing categorical values.

FR-06: PostgreSQL Storage

The system shall load raw and processed datasets into PostgreSQL data warehouse..

The database shall contain:

- Staging tables, Dimension tables, Curated datasets, Supporting metadata where required.

FR-07: Data Profiling

Python/Pandas shall be used to profile pharmaceutical datasets.

Profiling shall identify:

- Number of records, Number of columns, Missing values, Duplicate records, Data types, Unique values, Outliers, Inconsistent values.

FR-08: Data Lineage

The system shall document the movement of data from source to target.

The lineage documentation shall identify:

Source → Transformation → Target

For example:

Kaggle CSV → Pentaho Cleaning → Pentaho Transformation → PostgreSQL Staging → PostgreSQL Warehouse

FR-9: ETL Logging and Exception Handling

The ETL pipelines shall maintain information about:

- Successful records, Failed records, Transformation errors, Validation failures, Processing status

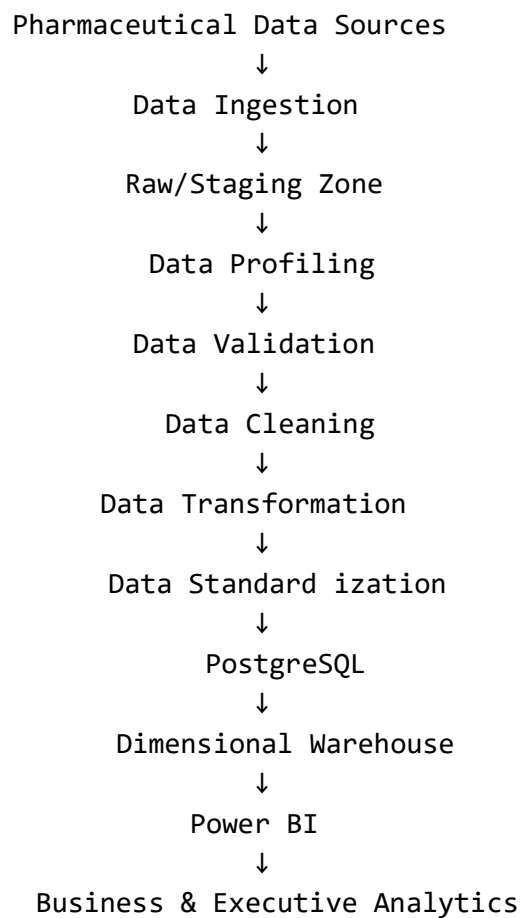
Failed records shall be identifiable for further investigation.

FR-10: Dashboard Development

Power BI shall be used to create dashboards from the curated PostgreSQL data.

8. Proposed ETL Workflow

The high-level ETL workflow is:



9. Technology Requirements

Category	Technology
ETL	Pentaho Data Integration (Spoon)
Database	PostgreSQL
Programming	Python / Pandas
Version Control	Git / GitHub

Reporting
Documentation
Project Management

Power BI
Markdown / MS Word
Agile Scrum

10. Constraints

- Source datasets may contain missing or inconsistent values.
- Different sources may use different formats and naming conventions.
- Data transformation rules may vary between source systems.
- The project is implemented as an academic enterprise data-engineering project rather than a production pharmaceutical platform.
- The quality of analytics depends on the quality and completeness of the available source datasets.

11. Conclusion

The Enterprise Pharmaceutical Data Platform will provide a unified approach for integrating heterogeneous pharmaceutical data sources, improving data quality, storing curated information in PostgreSQL, and enabling business analytics.

The proposed solution establishes the foundation for subsequent sprints involving data ingestion, profiling, warehouse development, governance, lineage, and analytics.